NATIONAL BIOINFORMATICS INFRASTRUCTURE SWEDEN

SciLifeLab

Introduction to Exercises

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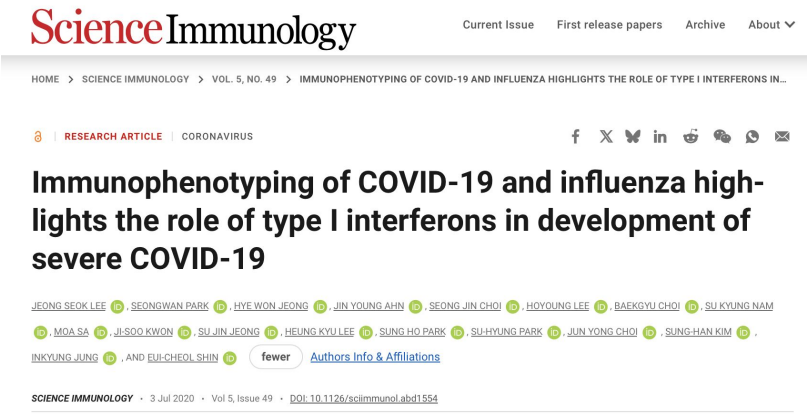
Outline

- Data set
- Analysis steps
- Toolkits and data structures
- Labs setup
- Practical information
- Demo

Data set



Science Immunology

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RESEARCH ARTICLE | CORONAVIRUS

Immunophenotyping of COVID-19 and influenza highlights the role of type I interferons in development of severe COVID-19

JEONG SEOK LEE, SEONGWAN PARK, HYE WON JEONG, JIN YOUNG AHN, SEONG JIN CHOI, HOYOUNG LEE, BAEKGYU CHOI, SU KYUNG NAM, MOA SA, JI-SOO KWON, SU JIN JEONG, HEUNG KYU LEE, SUNG HO PARK, SU-HYUNG PARK, JUN YONG CHOI, SUNG-HAN KIM, INKYUNG JUNG, AND EUI-CHEOL SHIN fewer Authors Info & Affiliations

SCIENCE IMMUNOLOGY • 3 Jul 2020 • Vol 5, Issue 49 • DOI: 10.1126/sciimmunol.abd1554

- Selected 4 controls and 4 severe COVID-19 cases
- Cells are subsampled to 1500 cells per subject
- Aim: Which cell types and genes are altered when comparing blood immune cells from healthy vs disease?
- Note: Trajectory lab has a different data set

https://nbisweden.github.io/workshop-scRNAseq/other/data.html

https://www.science.org/doi/10.1126/sciimmunol.abd1554

Overview analysis steps

Topic	Seurat	Bioconductor	Scanpy
1 Quality Control			
2 Dimensionality reduction			
3 Data integration			
4 Clustering			
5 Differential expression			
6 Celltype prediction			
7 Trajectory inference			

https://nbisweden.github.io/workshop-scRNAseq/home_contents.html

Three main toolkits

	Seurat v5	Bioconductor	scanpy
Language	R	R	Python
Data structure	`Seurat Object`	`SingleCellExperiment Object`	`AnnData`
Key features	+ Actively maintained (by Satija Lab) + Open source + Single R package + Specifically designed for single cell analysis (spatial and multimodal analysis)	+ Community-developed + Open source + Interoperability across single-cell related packages (Orchestrating Single-Cell Analysis (OSCA) guidelines) - Scalability	+ Community-developed (part of scverse project) + Open source + Interoperability with Python ML libraries + Scalable

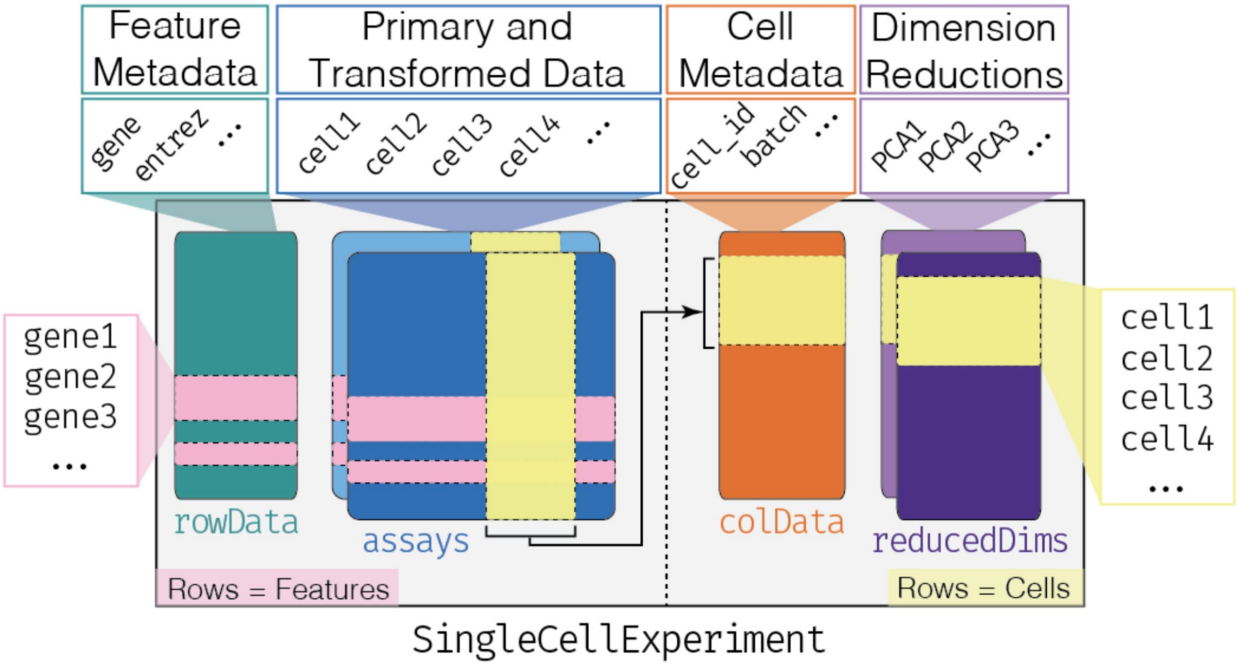
Seurat Object

```
▾ alldata
  ..@ assays: List of assays for count data
  .. ..$ RNA: Formal class 'Assay5'
  .. .. ..@ layers: List of data matrices (features x barcodes)
  .. .. .. .. ..$ counts: raw counts
  .. .. .. .. ..$ data: normalized counts
  .. .. .. .. ..$ scale: scaled counts
  ..@ meta.data: data.frame containing metadata for barcodes/cells
  ..@ active.assay: "RNA"
  ..@ reductions: List of dimensional reduction objects
  .. ..$ pca: (components x barcodes)
```

```
> slotNames(alldata)
[1]  "assays"    "meta.data"    "active.assay"    "active.ident"    "graphs"    "neighbors"
[7]  "reductions"    "images"    "project.name"    "misc"    "version"    "commands"
[13] "tools"
```

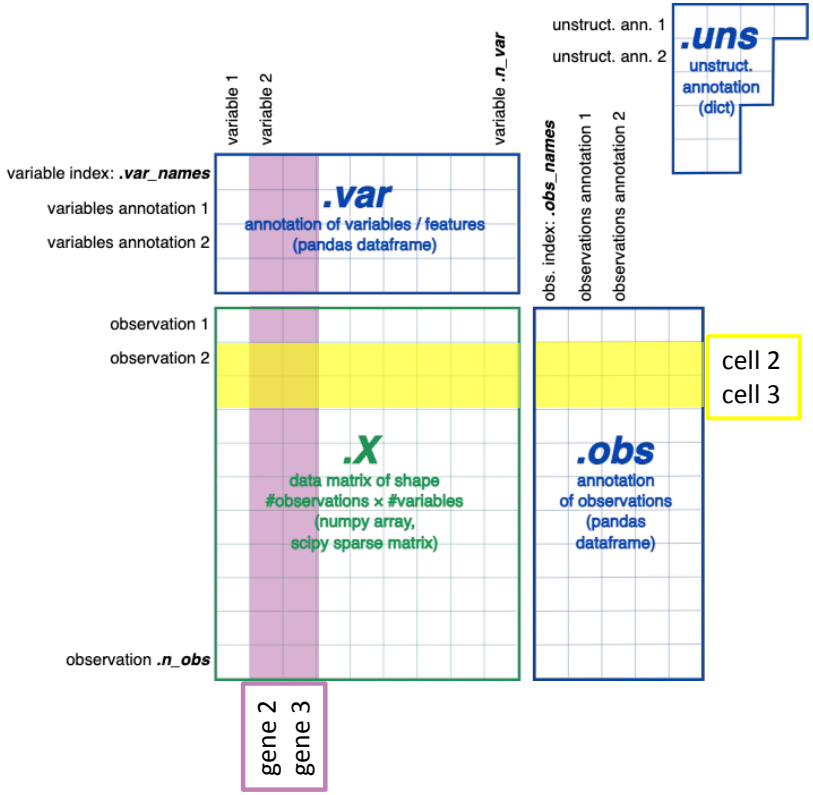
https://satijalab.org/seurat/articles/essential_commands.html

SingleCellExperiment Object



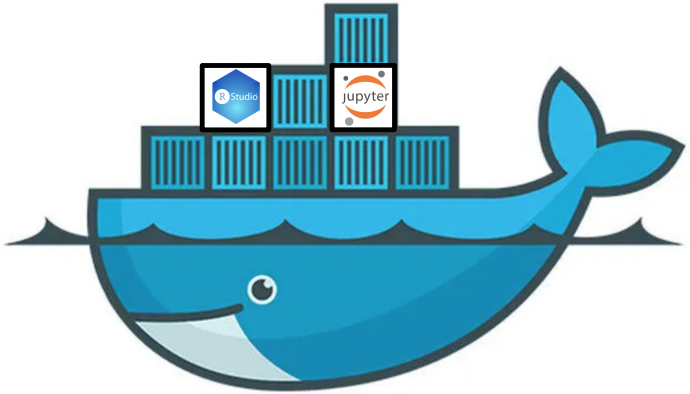
https://bioconductor.org/books/3.13/OSCA.intro/the-singlecellexperiment-class.html

AnnData Object



https://scanpy.readthedocs.io/en/stable/usage-principles.html

Containerized environments



- Two different environments
  - **RStudio** for Seurat and Bioconductor
  - **JupyterLab** for Scanpy
- The environments are provided as **Docker** images and are available on GitHub
- Choose a toolkit and start an instance on **SciLifeLab Serve** and access them through your browser

https://nbisweden.github.io/workshop-scRNAseq/other/containers.html

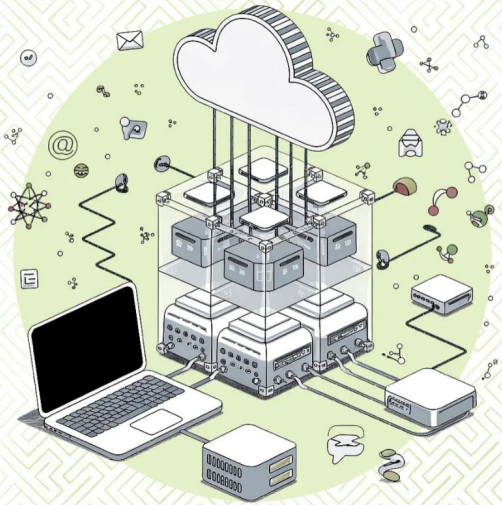
NBIS course: Tools for reproducible research

NBIS / Elixir workshop

Tools for Reproducible Research

How to work reproducibly with control and structuring of project code, environment and workflow management

Updated: 08-04-2026 at 11:55:57 .



https://nbisweden.github.io/workshop-reproducible-research

Running the exercises

- We provide scripts to download the exercises from GitHub
- The code is available as **Quarto notebooks** (`.qmd`) for Seurat and Bioconductor toolkits and as **Jupyter notebooks** (`.ipynb`) for Scanpy
- Please report errors in the code
  - Slack channel **#exercises**
  - The corrected code needs to be downloaded again using the download script

Getting help

- Ask for help in the Slack channel **#excercises**

- **Before asking**, please try to do some troubleshooting
  - Read the error messages carefully, most of the times the answer can be found there
  - Ask your colleagues in the breakout room
  - Check forums like SEQanswers, Stack Exchange, Bioconductor Forum or GitHub issues for the relevant package

Practical information

- Use the slide you created to introduce yourself in the breakout room
- Keep the camera on and discuss the exercises in the breakout room
- Fill in the quiz at the end of the day
- If you have time, experiment with different options in the algorithms

SciLifeLab Serve

Demo